Half-Wormholes as Wiener Chaos: Correlated-Replica Spectroscopy, Exact Pfaffian Sectors, and Boundary Moment Indeterminacy

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Abstract

The fixed-coupling partition function of one-time-point SYK is a homogeneous multilinear polynomial of degree $n = N/q$ in independent Gaussian couplings. Its square therefore has a canonical Wiener–Itô chaos decomposition. This is a probabilistic refinement of the exact half-wormhole expansion already constructed by Mukhametzhanov, not a replacement of that prior result. We derive five consequences of the canonical grading. First, two replicas with coupling correlation ρ obey $\mathbb{E}[z(J)z(J')] = \rho^n \mathbb{E}[z^2]$; scanning ρ diagonalises and measures the chaos-sector norms of z^2. Second, for $q = 2$ we reduce the previously known general variance sum to the closed formula $\mathbb{E}[W_{2k}^2] = (2n)!\binom{2(n-k)}{n-k}4^{-(n-k)}$. It yields an explicit correlated determinant formula for two Gaussian antisymmetric matrices. Third, the same profile gives $\mathbb{E}[z^4]/\mathbb{E}[z^2]^2 = 2n+1$ and quantifies the failure of a wormhole-plus-Wick-square truncation at $q = 2$. Fourth, Gaussian hypercontractivity supplies non-asymptotic, single-draw reconstruction bounds sector by sector. Fifth, the boundary moment problem for the law of z is determinate for $n \leq 2$ and indeterminate for every $n \geq 3$. We prove the latter by a conditional-Gaussian density lower bound and Krein's criterion. The result is a complete statement for the zero-dimensional model. It does not establish a dynamical bulk geometry or a complete theory of quantum gravity.

1 Introduction

In the one-time-point SYK model, fixed-coupling factorisation is represented by a finite polynomial identity. Saad, Shenker, Stanford and Yao identified the wormhole and coupling-sensitive half-wormhole saddles [13]. Mukhametzhanov subsequently constructed an exact finite expansion of z^2, including orthogonality, closed sector variances and the normal-ordered-square interpretation of the leading half-wormhole [9]. Related graded and Hermite structures appear in Tian and Yang [16]. Those results are prior art and are used here as such. After matching conventions, Mukhametzhanov's term Φ_{n-k} is precisely the chaos projection W_{2k} below. Thus neither the finite sector decomposition nor its general variance sum is claimed as new; the $q = 2$ binomial form is a closed reduction supplied with an independent generating-function proof.

Our starting point is narrower: over the Gaussian coupling ensemble, z^2 has a unique decomposition into eigenspaces of the Ornstein–Uhlenbeck (OU) semigroup. Identifying this canonical grading makes the correlated-replica kernel, Gaussian hypercontractivity and the Hamburger moment problem available without introducing new saddles. A targeted search recorded in the July 2026 source project found no substantive duplicate of the combined results in the half-wormhole, random-matrix and moment-problem literature. That historical search is bounded and has not been replaced by a comprehensive current novelty audit for this repository edition.

The exact contributions are:

1. the ρ^n two-replica identity and OU sector spectroscopy;

2. a closed $q = 2$ reduction of the known sector-variance sum, an independent enumeration, and the correlated antisymmetric-determinant corollary;

3. an explicit fourth-moment and truncation diagnosis, together with single-draw bounds;

4. the sharp determinacy dichotomy at chaos degree $n = 2/3$ and its spectral consequences.

We separate exact finite-N theorems from asymptotic interpretations. In particular, the standard Nualart–Peccati fourth-moment equivalence applies directly to sequences in a *fixed* Wiener chaos [11, 12]. In the conventional SYK scaling with fixed q and $N \to \infty$, the chaos order $n = N/q$ grows; fourth moments alone are then not advertised here as a central-limit proof. The L^2 truncation statements we use instead follow from the exact sector variances.

2 Model and canonical grading

Let $q \geq 2$ be even, $N = qn$, and let A run over the q-subsets of $[N]$. The couplings J_A are independent standard normal variables. Berezin integration gives, up to an irrelevant global phase,

$$z(J) = \sum_{P \in \mathrm{Part}_q(N)} \epsilon(P) \prod_{A \in P} J_A, \tag{1}$$

where P partitions $[N]$ into n blocks of size q and $\epsilon(P) = \pm 1$ is the Grassmann sign. For $q = 2$, $z(J) = \mathrm{Pf}\, J$ for the antisymmetric matrix with upper-triangular entries J_{ij}.

Let $L^2(\gamma) = \bigoplus_{d \geq 0} \mathcal{C}_d$ be the Wiener–Itô decomposition of the Gaussian coupling space [3, 4]. The OU operator is

$$(T_\rho F)(J) = \mathbb{E}_K F\Big(\rho J + \sqrt{1-\rho^2} K\Big), \qquad -1 \leq \rho \leq 1, \tag{2}$$

and acts on $\mathcal{C}_d$ by multiplication by ρ^d.

Proposition 2.1 (Purity and symmetry)**.** *$z \in \mathcal{C}_n$, $\mathbb{E}[z^2] = |\mathrm{Part}_q(N)|$, and the law of z is symmetric.*

Proof. Every monomial in (1) is a product of n distinct first Hermite polynomials, hence belongs to $\mathcal{C}_n$. Distinct partitions give distinct monomials, so orthogonality gives the second statement. Fix a vertex, say 1, and change the sign of every coupling J_A with $1 \in A$. Every partition contains exactly one such block, so $z \mapsto -z$, whereas the joint Gaussian law is unchanged. Thus all odd moments vanish and the law is symmetric. □

Theorem 2.2 (Correlated-replica identity)**.** *Let $J' = \rho J + \sqrt{1-\rho^2} K$ with K an independent coupling draw. Then*

$$\mathbb{E}[z(J) z(J')] = \rho^n \mathbb{E}[z^2]. \tag{3}$$

More generally, if $X = \sum_d X_d$ and $Y = \sum_d Y_d$ are their chaos decompositions, then

$$\mathbb{E}[X(J) Y(J')] = \sum_d \rho^d \mathbb{E}[X_d Y_d]. \tag{4}$$

Proof. Equation (4) is Mehler's identity $\mathbb{E}[X(J)Y(J')] = \langle X, T_\rho Y \rangle$. Proposition 2.1 then gives (3). □

Define the canonical half-wormhole sectors by

$$z^2 = \sum_{k=0}^{n} W_{2k}, \qquad W_{2k} \in \mathcal{C}_{2k}, \qquad \mathbb{E}[W_{2k} W_{2\ell}] = 0 \quad (k \neq \ell). \tag{5}$$

Thus $W_0 = \mathbb{E}[z^2]$, while $W_{2n} =: z^2 :$ is the Wick square. This decomposition is canonical because the $\mathcal{C}_d$ are the distinct OU eigenspaces. It need not coincide term-by-term with an arbitrary orthogonal expansion once $n \geq 3$. For the normal-ordered expansion of Mukhametzhanov [9], however, homogeneity gives the exact identification $\Phi_{n-k} = W_{2k}$ after matching the coupling normalisation.

Applying (4) to z^2 gives

$$\mathbb{E}[z^2(J)z^2(J')] = \sum_{k=0}^{n} \rho^{2k} \mathbb{E}[W_{2k}^2]. \tag{6}$$

Hence correlation in coupling space is a sector spectrometer: the coefficients of the polynomial in ρ^2 are precisely the sector norms.

3 Exact Pfaffian sector spectrum

For $q = 2$, ordered pairs of perfect matchings provide a closed enumeration.

Theorem 3.1 ($q = 2$ sector profile)**.** *For $z = \operatorname{Pf} J$ with J a $2n \times 2n$ Gaussian antisymmetric matrix,*

$$\boxed{\mathbb{E}[W_{2k}^2] = (2n)! \binom{2m}{m} 4^{-m}, \qquad m = n - k, \quad 0 \leq k \leq n.} \tag{7}$$

Equivalently,

$$\sum_{n \geq 0} \sum_{k=0}^{n} \mathbb{E}[W_{2k}^2] x^{2k} \frac{y^{2n}}{(2n)!} = \frac{1}{(1 - x^2 y^2)\sqrt{1 - y^2}}. \tag{8}$$

Proof. Expand z^2 over ordered perfect matchings (P, Q). Their symmetric difference is a vertex-disjoint union U of alternating even cycles of length at least four; shared edges contribute $J_e^2 = H_2(J_e) + 1$. A chaos monomial is therefore labelled by a set T of excited shared edges and by U, with order $2|T| + |E(U)|$. Each alternating cycle has two splittings into (P, Q), and the unused vertices carry an arbitrary common matching. Upon squaring a coefficient, the two common matchings form another union of doubled edges and alternating cycles. Sign coherence holds because matching edges have even block size.

The squared norm is thus the exponential generating function of labelled components of four types: an excited doubled edge of weight $2x^2$, a first-family cycle of length $2j \geq 4$ of weight $4x^{2j}$, an unexcited doubled edge of weight 1, and a second-family cycle of length $2j \geq 4$ of weight 2. Since the number of undirected alternating cycles on $2j$ labelled vertices supplies the standard $1/(2j)$ exponential weight, the exponential formula gives

$$\exp\left[x^2 y^2 + \sum_{j \geq 2} \frac{(xy)^{2j}}{j} + \frac{y^2}{2} + \sum_{j \geq 2} \frac{y^{2j}}{2j}\right] = \frac{1}{(1 - x^2 y^2)\sqrt{1 - y^2}}.$$

Extracting $x^{2k} y^{2n}$ and using $[u^m](1 - u)^{-1/2} = \binom{2m}{m} 4^{-m}$ proves (7). □

Equation (7) is the $q = 2$ evaluation of the general finite variance sum in Mukhametzhanov [9], not a priority claim for the sector variances. The proof above is an independent labelled-component derivation that exposes the rational generating function used in the following consequences.

Two checks are immediate. At $k = 0$, (7) equals $\mathbb{E}[z^2]^2 = ((2n - 1)!!)^2$. Summing over k and using

$$\sum_{m=0}^{n} \binom{2m}{m} 4^{-m} = (2n + 1) \binom{2n}{n} 4^{-n} \tag{9}$$

gives

$$\frac{\mathbb{E}[z^4]}{\mathbb{E}[z^2]^2} = 2n + 1. \tag{10}$$

Corollary 3.2 (Correlated antisymmetric determinants)**.** *If $J' = \rho J + \sqrt{1-\rho^2}K$ with J, K independent Gaussian antisymmetric $2n \times 2n$ matrices, then*

$$\mathbb{E}[\det J \, \det J'] = (2n)! \sum_{m=0}^{n} \binom{2m}{m} 4^{-m} \rho^{2(n-m)}. \tag{11}$$

Proof. $\det J = \mathrm{Pf}(J)^2 = z^2$. Combine (6) with Theorem 3.1. □

Formula (11) is distinct from the general characteristic-polynomial averages treated in random-matrix theory [2, 8]. We have not found this correlated antisymmetric specialisation stated in the searched literature. This negative search is recorded as evidence, not as a mathematical proof of priority.

4 Truncation and single-draw control

At $q = 2$, the fraction of the total L^2 norm in sector k is proportional to $\binom{2(n-k)}{n-k}4^{-(n-k)}$. The Wick-square fraction is

$$\frac{\mathbb{E}[W_{2n}^2]}{\mathbb{E}[z^4]} = \frac{1}{(2n+1)\binom{2n}{n}4^{-n}} \sim \frac{\sqrt{\pi}}{2\sqrt{n}}. \tag{12}$$

Thus no top sector dominates, and the wormhole-plus-Wick-square truncation fails in relative L^2 as $n \to \infty$. Equation (10) also shows divergence of the normalised fourth moment. This rules out Gaussian convergence in fourth moment; we do not infer a general weak-limit theorem from this fact alone.

For any sector, hypercontractivity converts its exact norm into a single-draw bound.

Lemma 4.1 (Chaos tail envelope)**.** *If $X \in \mathcal{C}_d$, then for $t > 0$*

$$\mathbb{P}(|X| \geq t\|X\|_2) \leq B_d(t) := \inf_{p \geq 2} (p-1)^{pd/2} t^{-p}. \tag{13}$$

In particular $B_d(t) \leq t^{-2}$, and for $t \geq (2e)^{d/2}$, $B_d(t) \leq \exp[-(d/2e)t^{2/d}]$.

Proof. Gaussian hypercontractivity gives $\|X\|_p \leq (p-1)^{d/2}\|X\|_2$ [4, 10]. Markov's inequality gives (13). The two displayed bounds follow from $p = 2$ and from the admissible choice $p = t^{2/d}/e$, respectively. □

Put $f_k = \mathbb{E}[W_{2k}^2]/\mathbb{E}[z^2]^2$ and $E_{\mathrm{tr}} = z^2 - \mathbb{E}[z^2] - W_{2n} = \sum_{k=1}^{n-1} W_{2k}$.

Theorem 4.2 (Single-draw reconstruction)**.** *For any $\varepsilon_k > 0$ with $\sum_{k=1}^{n-1} \varepsilon_k = \varepsilon$,*

$$\mathbb{P}\big(|E_{\mathrm{tr}}| \geq \varepsilon \mathbb{E}[z^2]\big) \leq \min\left\{ \frac{\sum_{k=1}^{n-1} f_k}{\varepsilon^2}, \sum_{k=1}^{n-1} B_{2k}\left(\frac{\varepsilon_k}{\sqrt{f_k}}\right) \right\}. \tag{14}$$

Proof. Orthogonality gives $\mathbb{E}[E_{\mathrm{tr}}^2]/\mathbb{E}[z^2]^2 = \sum_{k=1}^{n-1} f_k$, hence the first bound. For the second, if $|\sum W_{2k}| \geq \sum \varepsilon_k \mathbb{E}[z^2]$, at least one sector exceeds its threshold. Apply Lemma 4.1 and the union bound. □

Theorem 4.2 is non-asymptotic and valid for one fixed coupling draw. It does not assert that its dimension-free constants are optimal. When prior exact sector estimates show $\sum_{k=1}^{n-1} f_k \to 0$ in a specified large-N regime [9], the first term already proves relative L^2 reconstruction; the tail term refines atypical events sector by sector.

5 Boundary moment determinacy

Let μ be the law of z. In the baby-universe/GNS dictionary, the moments $\mu_j = \mathbb{E}[z^j]$ are multi-boundary data, and multiplication by z acts on the closure of polynomials in $L^2(\mu)$ [5, 7]. Whether all moments determine μ is exactly the Hamburger moment problem.

Theorem 5.1 (Determinacy dichotomy)**.** *For the one-time-point model* (1)*:*

1. *if $n \le 2$, the law μ is Hamburger determinate;*
2. *if $n \ge 3$, the law μ is Hamburger indeterminate.*

Proof. For $n \le 2$, hypercontractivity applied to $z \in \mathcal{C}_n$ gives

$$\mu_{2m}^{1/(2m)} = \|z\|_{2m} \le (2m-1)^{n/2}\|z\|_2.$$

Therefore $\sum_m \mu_{2m}^{-1/(2m)}$ dominates a constant multiple of $\sum_m m^{-n/2}$ and diverges. Carleman's criterion implies determinacy [1, 14].

Now let $n \ge 3$. Expand along blocks containing vertex 1:

$$z = \sum_{A \ni 1} \epsilon_A J_A C_A, \tag{15}$$

where every cofactor C_A uses only couplings on subsets not containing 1 and is independent of the family $\{J_A : A \ni 1\}$. Conditionally on the cofactors, $z \sim \mathcal{N}(0, S^2)$ with $S^2 = \sum_{A \ni 1} C_A^2$. Since the cofactor vector is not identically zero and its common zero set is algebraic of Gaussian measure zero, $S > 0$ almost surely. Thus μ has a strictly positive Gaussian-mixture density

$$f(t) = \mathbb{E}\left[(2\pi S^2)^{-1/2} \exp\left(-\frac{t^2}{2S^2}\right)\right]. \tag{16}$$

We need a lower tail bound on f. Fix a partition $\{A_0, B_2, \dots, B_n\}$ with $1 \in A_0$. On the event that $J_{B_j} \in [a, a+1]$ for $j = 2, \dots, n$ and all other couplings not containing 1 have absolute value at most one, the leading monomial of C_{A_0} has degree $n-1$. All its other monomials, and every C_A with $A \ne A_0$, contain at most $n-2$ of the selected large variables. Hence, for a above a model-dependent constant,

$$c a^{n-1} \le S \le C a^{n-1}, \qquad \mathbb{P}(E_a) \ge c_1 e^{-c_2 a^2}.$$

Insert this event into (16) and choose $a = t^{1/n}$. For sufficiently large t,

$$f(t) \ge c_3 t^{-(n-1)/n} \exp[-c_4 t^{2/n}]. \tag{17}$$

Because $2/n < 1$, (17) implies

$$\int_{\mathbb{R}} \frac{-\log f(t)}{1+t^2}\, dt < \infty$$

up to an irrelevant finite contribution on compact intervals. Krein's criterion therefore implies Hamburger indeterminacy [6, 15]. This proves the dichotomy. □

For $n \ge 3$, the symmetric Jacobi operator associated with the boundary moments is not essentially self-adjoint on polynomials; its canonical self-adjoint extensions form the Nevanlinna family and have discrete extremal spectral measures [1, 14]. The physical coupling law (16) is continuous and non-extremal, so polynomials are not dense in its full L^2 space. This does not make the GNS construction inconsistent. It says only that one-time boundary moments do not select the microscopic coupling measure uniquely once one boundary amplitude is a Gaussian polynomial of degree at least three.

6 Scope and conclusion

The exact half-wormhole expansion, its identification with the chaos projections, and its general sector-variance sum are not new. Our contribution is the correlated-replica use of the OU grading and the additional consequences developed here. Equations (3) and (6) turn correlated replicas into a sector probe. Theorem 3.1 gives a closed Pfaffian reduction and independent proof, while Corollary 3.2 gives the correlated-determinant consequence. Theorem 4.2 is a finite single-draw statement. Theorem 5.1 sharply separates determinate and indeterminate boundary laws.

The limitations are equally sharp. The model is zero-dimensional; "wormhole" labels the ensemble-connected term, not a derived spacetime geometry. Gaussianity is essential to the OU and hypercontractive form used here. A thermal partition function has an infinite chaos expansion. Finally, when the chaos order grows with N, the fixed-chaos fourth-moment theorem cannot by itself prove a central limit. These boundaries prevent the manuscript from being read as a construction of complete quantum gravity; it is a rigorous result about a solvable factorisation model.

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